

Hallucination Detection and Mitigation in LLMs

Final Year Project Presentation

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| Background and Problem Scope

- ✓ **Defining Hallucination:** Model outputs that are nonsensical or unfaithful to the source text/world knowledge.
- ✓ **The Trust Gap:** Inaccurate information erodes reliability in mission-critical applications.
- ✓ **QA System Impact:** Factual consistency is paramount for legal, medical, and technical assistants.
- ✓ **Scale of Issue:** Even state-of-the-art models exhibit up to 15-20% hallucination rates in complex tasks.

Motivation for Research

- ✓ **Limitations of Current Methods:** Existing tools often separate detection from mitigation stages.
- ✓ **Single-Signal Constraints:** Heavy reliance on NLI (Natural Language Inference) lacks semantic depth.
- ✓ **Unified Framework Need:** A real-time system that detects and corrects in a single pipeline is required.
- ✓ **Efficiency:** Balancing detection accuracy with computational latency for production use.

| Project Aim & Objectives

- ✓ **Primary Goal:** Build a multi-signal framework for robust hallucination detection.
- ✓ **Objective 1:** Integrate "LLM-as-a-Judge" and NLI signals for high-precision verification.
- ✓ **Objective 2:** Implement a tiered mitigation logic (Keep, Rewrite, Correct).
- ✓ **Objective 3:** Validate results across standard benchmarks (FEVER, NQ, TruthfulQA).

Project Implementation Timeline

Review



Literature review and formulation of the multi-signal framework.

Design



Architecture design and selection of NLI/Judge models.

Implementation



Python-based pipeline using Ollama and HuggingFace.

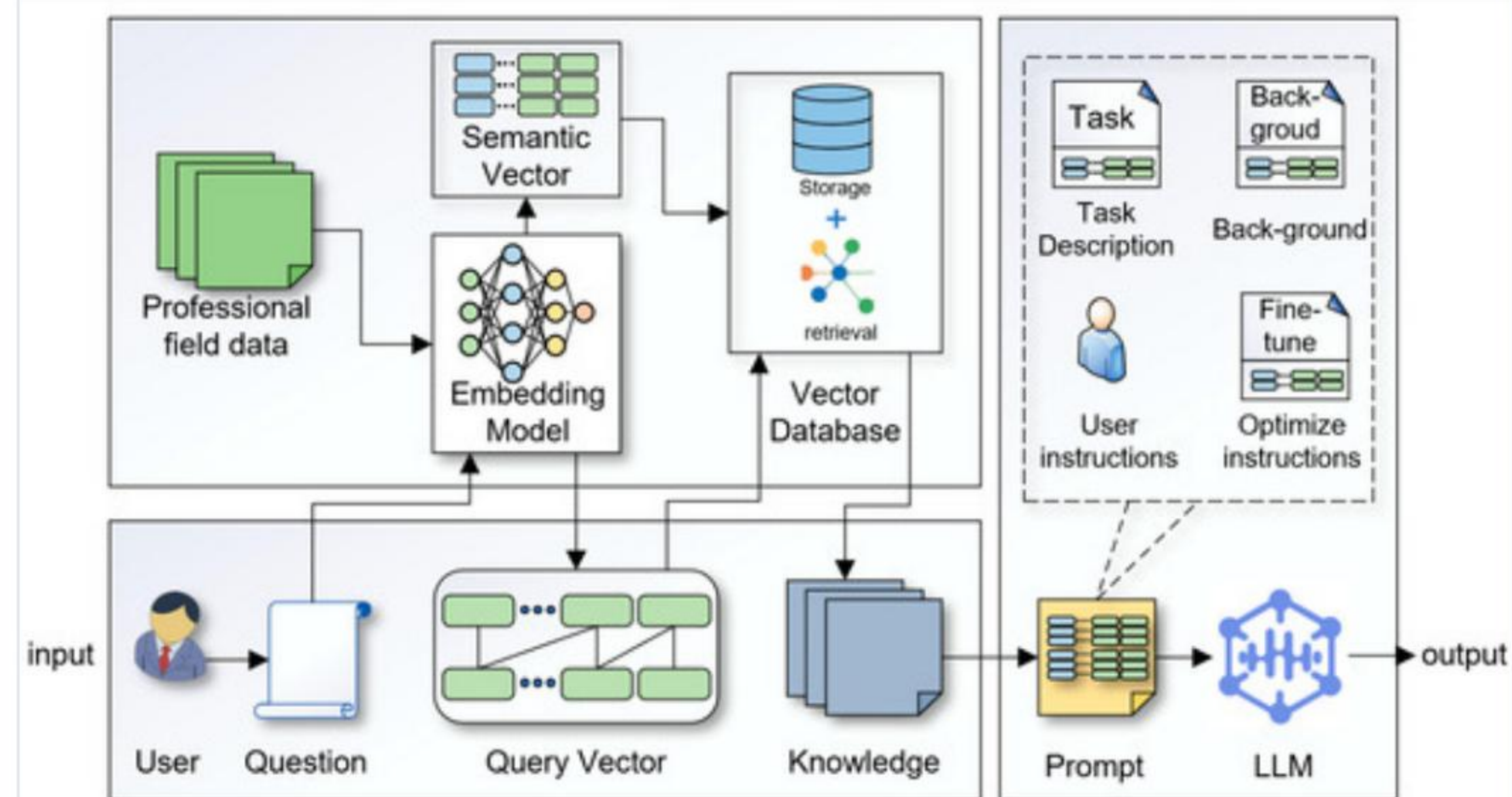
Evaluation



Performance analysis and hallucination rate calculation.

System Architecture Overview

- ✓ **Pipeline Flow:** Input → LLM Answer → Multi-Signal Detection → Mitigation Policy → Result.
- ✓ **LLM Inference:** Utilizes *Ollama* with Llama2 for generation and Qwen for judging.
- ✓ **Grounding:** NLI models used for factual consistency and entailment checks.



Multi-Signal Detection Framework



LLM-as-a-Judge

Semantic evaluation using Qwen to identify internal contradictions in the output.



NLI Verification

Natural Language Inference models ensure the claim is entailed by the provided source context.



Self-Consistency

Measuring variance across multiple inference samples to identify "uncertain" hallucinations.



Reliability Score

A weighted combination of signals producing a final confidence index between 0 and 1.

Tiered Mitigation Strategy

Score (t)	Risk Level	System Action
$t < 0.3$	Low Risk	Accept: Retain original answer as factually sound.
$0.3 \leq t < 0.7$	Medium Risk	Rewrite: Prompt model to revise and clarify the output.
$t \geq 0.7$	High Risk	Correct: Strong intervention or rejection of the answer.

Policy: Always select the candidate answer with the absolute minimum hallucination score.

Key Performance Results



Performance variance highlights the challenge of balancing grounding (FEVER) vs. open-domain accuracy (TruthfulQA).

Conclusion & Future Work

Key Takeaways

- ✓ Multi-signal detection outperforms single-model judging.
- ✓ Significant reductions in hallucinations for grounded verification tasks.
- ✓ Tiered mitigation ensures system efficiency.

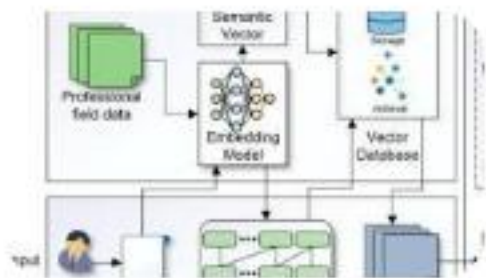
Next Steps

- ✓ Integrate **RAG** for real-time external grounding.
- ✓ Optimize Judge models for lower latency.
- ✓ Expand testing to multi-lingual scenarios.

Questions?

Contact: student@university.edu

Image Sources



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